

Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

Project Documents

8 encoders | 9 findings | 11 falsified explanations | 2 shipped systems

EXECUTIVE FINDING

Models do not share raw coordinates or one rigid geometry. They preserve relational structure, and a fitted linear map exposes enough for practical transfer. This is weak, constructive convergence - not a universal representation.

RESEARCH PROGRAM

1

Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R² mean 0.737; peak 0.797

2

Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.

R@5 = 95.3% of native ceiling

3

Cheap Adapter hub

Seven experts, one space

One entry map per encoder reaches a shared 512-d whitened hub, replacing quadratic pairwise translation with linear growth.

N maps, not N(N-1)

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

1	Shared neighbours	mutual k-NN	strong locally
2	Shared order	distance-rank rho	0.29 cross-modal
3	Shared structure	CKA / RSA	partial
4	Shared subspace	ridge + retrieval	88-97% of ceiling
5	Same geometry	rotation only	rejected: 0.038

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.9183

MLP and linear map tie exactly

0.592

vision adapter held-out R²

0.768

bge-SBERT raw shape rho

96.7%

ConvNeXt hub transfer / native

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. Cheap Adapter asks what is shared, whether one matrix makes it useful, and where the method breaks: mainly local structure; transfer reaches 96.7% of native on held-out ConvNeXt. The source-rank cliff is numerical, not a representational boundary, and this does not establish one universal representation.

From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

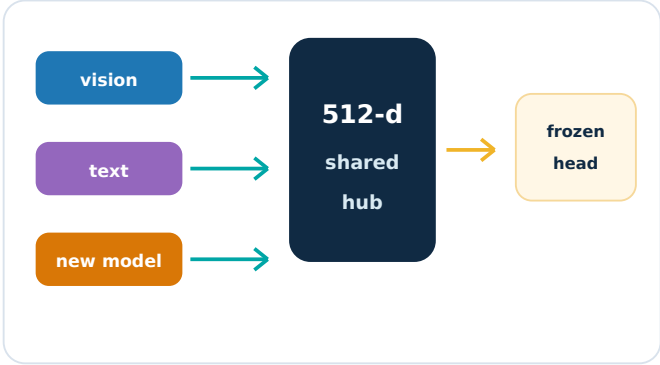
PAIRWISE

7 encoders -> 42 ordered maps

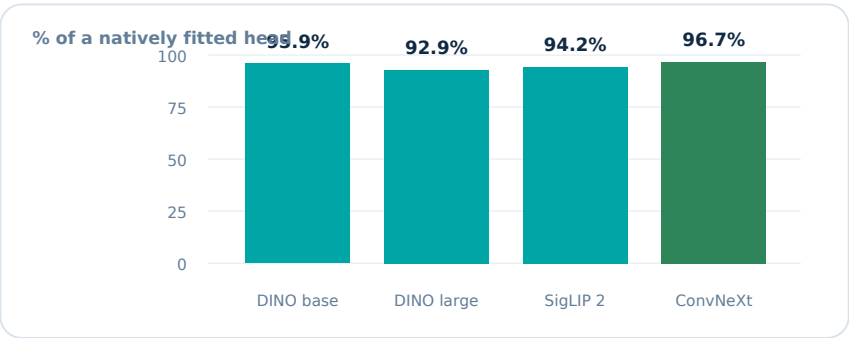
Adding one model requires maps to and from every existing model.

CHEAP ADAPTER HUB

7 encoders -> 7 entry maps



MEASURED TRANSFER - HELD-OUT 1,000-ITEM RETRIEVAL



WHAT TRANSFERRED

- A caption head trained on DINOv2-small serves encoders it never saw.
- SigLIP 2 tests independent lineage; ConvNeXt tests a different architecture.
- A new encoder still needs a fitted entry map: this is adaptation, not zero-shot.

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.768.

COMPOSED

103.6%

Mean hub performance versus bespoke pairwise maps across all 42 ordered pairs.

DEPLOYED

0.79 MB

The fp16 MobileCLIP-to-SigLIP matrix reaches 95.3% of native R@5.

WHY THE INFERENCE IS CREDIBLE

Frozen models only closed-form maps are fitted	Held-out scoring 8,533 train / 1,000 test in the hub	Negative controls shuffle and random maps sit at chance	Nonlinear control MLP ties linear at 0.9183
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PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local - nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

Fractal probe: the role of training data

FractalDB-only encoders provide a sharp control: structured training with zero natural photographs, bracketed by matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Fractal vs controls

Fractal training adds a small signal above architecture alone, but remains far below natural-image training. Image-vs-text reality dependence is inconclusive.

G10

Functional recovery

A fitted Cheap Adapter entry map plus the frozen caption head finds no FractalDB-specific retrieval advantage over random initialization.

G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN control: 0.067 fractal vs 0.036 random. The training effect survives preprocessing correction.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

ENCODER	TRANSFER	NATIVE	RANDOM TWIN	INTERPRETATION
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	-	clear functional transfer
ImageNet ViT	0.204	0.208	-	clear functional transfer

FINAL CONCLUSIONS

1

Convergence is real but bounded.

The tested models share local relational structure and useful linearly recoverable subspaces, not one exact global representation.

2

Linear maps are functionally sufficient here.

Nonlinear capacity adds no measurable gain; rotation alone fails because anisotropic stretch matters.

3

Natural-world training is the decisive source.

Fractals add a small visible geometric signal, but not a FractalDB-specific semantic transfer advantage through Cheap Adapter.

4

Geometry and utility must be tested separately.

Low raw agreement can coexist with high transfer, while floor-level native retrieval makes transfer ratios meaningless.

SCOPE

Results are bounded to the tested datasets, pooling choices, widths, and linear readouts. Held-out architectures receive fitted entry maps; fractal conclusions cover two released checkpoints; direct CNN-vs-ViT comparison is capacity-confounded.